



Monolith

M I C R O S Y S T E M S

EMBEDDED SYSTEMS ENGINEERING

Industrial Automation · Autonomous Systems · Edge AI · IIoT

14+

Years Experience

25+

Projects Delivered

500+

AGVs Deployed

1,000+

Devices Running

ISRO

Verified Experience

DOMAINS

Industrial Automation

Autonomous Systems

Satellite Comms

Edge AI

Medical

Electric Mobility

COMPANY

Monolith Microsystems Private Limited

Incorporated under Companies Act, 2013

CONTACT

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Ahmedabad, India · Available Globally

This document is confidential and intended for the recipient only.

Company Overview

ABOUT MONOLITH MICROSYSTEMS

Monolith Microsystems is a specialized embedded systems engineering firm founded by Milan Panchal — an engineer with 14+ years of hands-on experience spanning industrial automation, autonomous systems, satellite communication, electric mobility, and medical infrastructure.

Most embedded projects fail during the transition from prototype to production. Monolith bridges that gap by combining architecture depth, firmware expertise, hardware integration, industrial networking, manufacturing awareness, and end-to-end project execution — all under direct founder involvement.

*"Engineering should solve real problems.
Technology must survive outside the lab."*

— Milan Panchal, Founder

WHAT MAKES US DIFFERENT

01 Architecture Thinking

Every engagement starts with system design — not code. Built for scale, reliability, and longevity from the first conversation.

02 Production Mindset

DFM awareness, validation strategy, and manufacturability baked in from Day 1. Prototypes are not the goal — shipped products are.

03 Cross-Domain Expertise

Firmware, hardware, industrial networking, edge AI, IoT, and cloud — one team across the full embedded stack.

04 Direct Founder Involvement

Milan Panchal is hands-on on every project. No account managers, no handoffs. You work directly with the expert.

KEY METRICS

14+

Years Experience

25+

Projects Delivered

500+

AGVs Deployed

1,000+

Devices Running

30+

Engineers Led

ISROSpace-grade
Experience

Solutions

ENGINEERING SOLUTIONS

Monolith delivers complete embedded product solutions across six primary domains. Each solution spans from architecture through firmware, hardware, testing, and production support.

Autonomous Systems

AGV/AMR platforms, computer vision pipelines, LiDAR integration, and autonomous navigation stacks. 500+ units deployed across Indian manufacturing facilities.

AGV AMR LiDAR OpenCV Navigation

Industrial Automation

PLC systems, control room infrastructure, industrial lighting control (DMX/DALI), CAN networks, and SCADA integration.

PLC CAN DALI DMX SCADA Modbus

Satellite & Remote Connectivity

Satellite IoT gateways and remote asset monitoring platforms for mining, defense, utilities, and agriculture. ISRO-grade engineering.

ISRO Satellite IoT LoRa Gateway Remote Assets

Edge AI Systems

On-device inference, computer vision pipelines, LiDAR processing, and voice interface integration for industrial environments.

OpenCV Inference Vision AI LiDAR Voice

Medical Infrastructure

Operation theatre monitoring systems, gas distribution control, and environmental sensing for healthcare facilities.

OT Systems Gas Monitoring Env. Sensor Modular

Electric Mobility

Battery management systems, motor controllers, and fleet management solutions for e-bikes and EV platforms.

BMS Motor Control Fleet Mgmt E-bike EV

PROJECT HIGHLIGHTS

500+ AGVs Deployed Largest AGV deployment across Indian manufacturing sites.	1,000+ Devices Running Industrial IoT network running continuously in production.	Satellite IoT (ISRO) Remote monitoring gateway, space-program quality.	Smart OT Platform Modular OT control: gas, lighting, environment unified.
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Capabilities

THE COMPLETE EMBEDDED STACK

From bare-metal firmware to cloud dashboards — Monolith covers every layer of the embedded product stack. This end-to-end capability eliminates integration risk and reduces handoff delays common in multi-vendor engagements.

EMBEDDED SOFTWARE — Architecture Design — RTOS / Bare-metal — Firmware Development — Device Management — OTA Updates	HARDWARE DESIGN — PCB Schematic & Layout — DFM Review — Validation & Bring-up — Component Sourcing	INDUSTRIAL NETWORKING — BLE / LoRa — CAN / Modbus / LinBus — DALI / DMX — MQTT / CoAP
EDGE AI — Computer Vision (OpenCV) — LiDAR Processing — On-device Inference — Voice Interfaces	IOT ARCHITECTURE — Node Design — Gateway Integration — Cloud Connectivity — Dashboard & Analytics	PRODUCT LIFECYCLE — Concept & Architecture — Prototype → Validation — Production Readiness — Long-term Support

PRODUCT DEVELOPMENT LIFECYCLE

01 Concept	02 Architecture	03 Prototype	04 Validation	05 Production	06 Support
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About & Contact

FOUNDER — MILAN PANCHAL

**14+**
Years**25+**
Projects**30+**
Engineers Led**ISRO**
Experience

Milan Panchal

Embedded Systems Architect · Project Leader · Technology Strategist

14+ years of embedded engineering experience — from network infrastructure through ISRO-grade satellite communication to leading cross-functional teams of 30+ engineers on complex IIoT deployments.

Monolith Microsystems is the culmination of that journey: a focused practice that takes ambitious embedded hardware projects all the way from concept to production.

CAREER JOURNEY

Network Engineer	Embedded Engineer	ISRO Experience	Tech Lead → PM	Founder, Monolith
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INDUSTRY COVERAGE

Manufacturing Industry 4.0, Machine Monitoring, Predictive Maintenance	Logistics AGV Fleets, Asset Tracking, Warehouse Automation	Healthcare OT Monitoring, Gas Systems, Medical Infrastructure
Energy Power Monitoring, Smart Metering, Grid Edge Devices	Agriculture Remote Monitoring, Field Automation, IoT Sensors	Aerospace Satellite Communication, Telemetry, Ground Systems

CONTACT & SERVICE AREAS

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Let's Build Something That Ships.

Monolith Microsystems — where engineering discipline meets production reality.

THREE REASONS TO WORK WITH MONOLITH

01 We Architect First

Before a single line of code is written, the system architecture is designed for scalability, reliability, and long-term maintainability. This is the single biggest difference between prototypes that die in the lab and products that survive in the field.

02 We Ship Products

Our production mindset means DFM, validation, manufacturing awareness, and supply chain considerations are part of every design decision — not afterthoughts added when it is too late to change anything.

03 You Work With Milan

Direct founder involvement on every project.
14+ years of experience, ISRO credentials,
and a proven track record of 500+ deployed
AGVs and 1,000+ live industrial devices.

TECHNICAL DOMAINS

Embedded C/C++ · FreeRTOS · Zephyr · Linux BSP · PCB Design (KiCad/Altium) ·
Zigbee / BLE 5.x / LoRaWAN · CAN Bus / Modbus / LinBus · DALI / DMX512 · OpenCV /
TensorFlow Lite · LiDAR SLAM · AWS IoT / Azure IoT Hub · MQTT / CoAP / OTA

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